1.2.2 Use of apparatus and techniques

	Learning outcomes	Additional guidance
	*Through use of the apparatus and techniques listed below, and a minimum of 12 assessed practicals (see Section 5), learners should be able to demonstrate all of the practical skills listed within 1.2.1 and CPAC (Section 5, **Table 2**) as exemplified through:*	
(a)	use of appropriate apparatus to record a range of measurements (to include mass, time, volume of liquids and gases, temperature)	HSW4
(b)	use of a water bath or electric heater or sand bath for heating	HSW4
(c)	measurement of pH using pH charts, or pH meter, or pH probe on a data logger	HSW4
(d)	use of laboratory apparatus for a variety of experimental techniques including: **(i)** titration, using burette and pipette **(ii)** distillation and heating under reflux, including setting up glassware using retort stand and clamps **(iii)** qualitative tests for ions and organic functional groups **(iv)** filtration, including use of fluted filter paper, or filtration under reduced pressure	HSW4
(e)	use of a volumetric flask, including accurate technique for making up a standard solution	HSW4
(f)	use of acid–base indicators in titrations of weak/strong acids with weak/strong alkalis	HSW4
(g)	purification of: **(i)** a solid product by recrystallisation **(ii)** a liquid product, including use of a separating funnel	HSW4
(h)	use of melting point apparatus	HSW4
(i)	use of thin layer or paper chromatography	HSW4

(j)	setting up of electrochemical cells and measuring voltages	HSW4
(k)	safely and carefully handling solids and liquids, including corrosive, irritant, flammable and toxic substances	HSW4
(l)	measurement of rates of reaction by at least two different methods, for example: **(i)** an initial rate method such as a clock reaction **(ii)** a continuous monitoring method.	HSW4